

Elec-S1 · Part 1

Answer keys and marking schemes

Progress Tests 1.A → 3.C
Semester 1 · Academic Year 2026–2027

How to mark yourself

- Model answers are in **blue**, inside the same frames as on the paper. Your wording may differ; what matters is the content listed in the *marking scheme* below each exercise.
- Each circled item is worth the points shown. A numerical result without its unit, or with a wrong unit symbol, does not earn its item.
- Tick-box questions: 1 mark each. The “Why” line explains the answer and the trap behind the most tempting wrong option.
- Red boxes (*Examiner’s tip*) flag common traps, including the ones that cost marks in Test 1.

Lessons from Test 1 (21 September 2026)

Exercise	Score	What cost marks	Rule from now on
1. Unit conversion	1/1	“KV”: capital K is the kelvin.	Copy prefix symbols exactly: k (kilo), M (mega), m (milli), μ (micro).
2. Electric current	4/5	Unit of current given as “C s ⁻¹ ”; charge written “coulombs (s)”.	Name <i>and</i> symbol: ampere (A), a base unit; coulomb (C) = A s; second (s).
3. Circuit diagram	0/3	Counted the drawn junction dots (4 nodes); ignored the wires (4 components); 6 branches.	Nodes are <i>electrically distinct</i> (2); each wire segment between two dots is a component (4 + 6 = 10); branches join distinct nodes (4).
4. Drift velocity	3/5	Numbers put in before isolating v ; units missing; $1/(3.2 \times 10^4)$ turned into 3.2×10^{-3} .	$v = i/(neS)$ first, then numbers with units, powers of ten on their own: $v = 3.1 \times 10^{-5} \text{ m s}^{-1}$.

Counting convention used throughout. This follows the marking of Test 1. Points joined only by wires form a single node, and a node must join at least three conductors (a point between two components in series is not a node). A branch runs from one node to another. Independent node equations: $N - 1$. Independent meshes: $B - N + 1$.

Progress Test 1.A — Elec-S1

1.A · From potential to electric voltage

Semester 1 · Academic Year 2026–2027

ANSWER KEY · MODEL ANSWERS AND MARKING SCHEME

Duration: 7 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Unit conversion

(1 mark)

Convert the voltage $u = 0.0258$ MV into millivolts. Give the exact result in scientific notation, without rounding.

$u = 0.0258 \times 10^6 \text{ V} = 2.58 \times 10^4 \text{ V} = 2.58 \times 10^4 \times 10^3 \text{ mV}$
$u = 2.58 \times 10^7 \text{ mV}$

Marking

①
0 1
☐ ☐

Marking scheme

① **1 pt** 2.58×10^7 mV, exact, with the unit symbol written correctly.

Examiner's tip

M (mega, 10^6) and m (milli, 10^{-3}) differ only by their case, exactly like k (kilo) and K (kelvin) in Test 1. Go through the base unit: MV $\rightarrow$ V $\rightarrow$ mV.

Exercise 2. Electric voltage — definition

(5 marks)

Define the electric voltage u_{AB} between two points A and B of a circuit. Write down its expression in terms of the electric potentials, deduce the relation between u_{AB} and u_{BA} , and state the SI unit of voltage (name and symbol).

The voltage u_{AB} is the difference of electric potential between A and B. Only potential differences can be measured (like a difference in height: a height needs a reference, a height difference does not). $u_{AB} = V_A - V_B$ hence $u_{BA} = V_B - V_A = -(V_A - V_B) = -u_{AB}$. Voltage is a signed quantity: $u_{AB} > 0 \iff V_A > V_B$. SI unit: the volt, symbol V (same unit as the potential).

Marking

①
0 1 2
☐ ☐ ☐

②
0 1 2
☐ ☐ ☐

③
0 1
☐ ☐

Marking scheme

- ① **2 pts** Voltage = potential difference between the two points (1 pt only if the potentials are not mentioned).
- ② **2 pts** $u_{AB} = V_A - V_B$ (1 pt) and $u_{BA} = -u_{AB}$ (1 pt).
- ③ **1 pt** Name *and* symbol: volt, V.

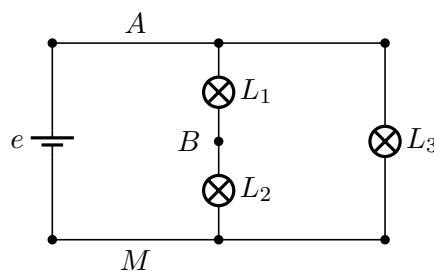
Examiner's tip

In Test 1 the unit mark was lost on a symbol. Always give both the name and the exact symbol of a unit: "volt (V)".

Exercise 3. Potentials in a circuit

(3 marks)

The circuit below contains an ideal voltage source of electromotive force $e = 6.0 \text{ V}$ and three lamps. The wires are ideal. The wire M , connected to the $-$ terminal of the source, is taken as the reference of potentials: $V_M = 0$.



How many distinct electric potentials are there in this circuit?

- ☐ 2
 ☒ 3
 ☐ 4
 ☐ 7

Why: an ideal wire has the same potential at every point, so count the electrically distinct wires: A (top), B (between L_1 and L_2) and M (bottom). The dots along one wire do not create new potentials.

The voltage across L_2 is $u_{BM} = 2.0 \text{ V}$. The voltage u_{AB} across L_1 is:

- ☐ 2.0 V
 ☒ 4.0 V
 ☐ 6.0 V
 ☐ 8.0 V

Why: $V_A = e = 6.0 \text{ V}$ and $V_B = u_{BM} = 2.0 \text{ V}$, so $u_{AB} = V_A - V_B = 4.0 \text{ V}$.

A voltmeter has its V terminal on B and its COM terminal on A . It displays:

- ☐ +4.0 V
 ☒ -4.0 V
 ☐ +2.0 V
 ☐ -6.0 V

Why: a voltmeter displays $V_V - V_{\text{COM}} = V_B - V_A = u_{BA} = -u_{AB}$. The sign is negative because COM is not on the side of the $-$ terminal.

(5 marks)

$V_A = 12.0 \text{ V}$	$V_B = 4.5 \text{ V}$	$V_C = -3.0 \text{ V}$
------------------------	-----------------------	------------------------

$u_{AB} = V_A - V_B = 12.0 \text{ V} - 4.5 \text{ V} = 7.5 \text{ V}$ $u_{BC} = V_B - V_C = 4.5 \text{ V} - (-3.0 \text{ V}) = 7.5 \text{ V}$ $u_{CA} = V_C - V_A = -3.0 \text{ V} - 12.0 \text{ V} = -15.0 \text{ V}$ $u_{AB} + u_{BC} + u_{CA} = (V_A - V_B) + (V_B - V_C) + (V_C - V_A) = 0$ (check: $7.5 \text{ V} + 7.5 \text{ V} - 15.0 \text{ V} = 0$). Going round the closed path $A \rightarrow B \rightarrow C \rightarrow A$ brings you back to the starting potential, like walking round a mountain back to your starting height. The result does not depend on the reference chosen. (This is the mesh law of section 3.A.)
--

1

01

2

01

3

01

4

01

5

01

- ① **1 pt** Literal expressions $u_{XY} = V_X - V_Y$ written before any number.
- ② **1 pt** $u_{AB} = 7.5 \text{ V}$.
- ③ **1 pt** $u_{BC} = 7.5 \text{ V}$ (negative potential handled correctly).
- ④ **1 pt** $u_{CA} = -15.0 \text{ V}$ (the sign is part of the answer).
- ⑤ **1 pt** Sum = 0, justified (potentials cancel two by two / closed path).

Test 1, Ex. 4: “Never mix figures and letters in the same equation.” Write the literal relation first, then substitute *with units*. Brackets around a negative value, $4.5\text{ V} - (-3.0\text{ V})$, prevent the classic sign slip.

Unofficial practice material written from the Elec-S1 Part 1 lecture notes · not an official examination paper

Progress Test 1.B — Elec-S1

1.B · From charge to electric current

Semester 1 · Academic Year 2026–2027

ANSWER KEY · MODEL ANSWERS AND MARKING SCHEME

Duration: 7 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Unit conversion

(1 mark)

The capacity of a phone battery is $Q = 4000 \text{ mA h}$, i.e. the charge it delivers when a current of 4000 mA flows for one hour. Express Q in coulombs. Give the exact result.

$Q = 4000 \times 10^{-3} \text{ A} \times 3600 \text{ s} = 4.000 \text{ A} \times 3600 \text{ s}$	(and $1 \text{ A s} = 1 \text{ C}$)
$Q = 14\,400 \text{ C} = 1.44 \times 10^4 \text{ C}$	

Marking

①
0 1
☐ ☐

Marking scheme

① 1 pt $1.44 \times 10^4 \text{ C}$ (the hour converted into 3600 s).

Examiner's tip

The coulomb is a *derived* unit, $1 \text{ C} = 1 \text{ A s}$: a charge is a current multiplied by a time. That is why a “mA·h” is a unit of charge.

Exercise 2. Microscopic expression of the current

(5 marks)

A conductor of cross-sectional area S contains n mobile charge carriers per unit volume, each of charge q , all moving with the drift velocity v along the axis of the conductor. Express the charge dq that crosses S during dt (explain your reasoning), deduce the current i , and check that your result is dimensionally consistent, stating the SI unit of n .

During dt each carrier moves by $dx = v dt$. The carriers that cross S are those contained in the cylinder of base S and length $v dt$, of volume $S v dt$:

$$dN = n S v dt \quad \Rightarrow \quad dq = q dN = n q S v dt$$

$$i = \frac{dq}{dt} = n q v S$$

(for electrons, $q = -e$: $i = -nevS$)

Units: n in m^{-3} ; $[n q v S] = \text{m}^{-3} \times \text{C} \times \text{m s}^{-1} \times \text{m}^2 = \text{C s}^{-1} = \text{A}$ (ampere). ✓

Marking

①		
0	1	2

②		
0	1	2

③	
0	1

Marking scheme

- ① **2 pts** Cylinder of volume $S v dt$ (1 pt) $\Rightarrow dq = n q S v dt$ (1 pt).
 ② **2 pts** $i = dq/dt$ (1 pt) $= n q v S$ (1 pt).
 ③ **1 pt** n in m^{-3} and $n q v S$ in $\text{C s}^{-1} = \text{A}$.

Examiner's tip

Test 1, Ex. 2: "current = C s^{-1} " did not earn the unit mark. The expected answer was the **ampere (A)**, one of the seven SI base units. Writing $\text{C s}^{-1} = \text{A}$ is a good dimensional check, but it does not name the unit.

Exercise 3. Current and ammeters

(3 marks)

In a copper wire carrying a current, the conduction electrons move:

- ☐ in the conventional direction of the current;
☒ in the direction opposite to the conventional direction of the current;
☐ at a speed close to that of the electric signal;
☐ only when the wire is heated.

Why: the conventional direction is that of positive charges. Electrons carry $-e$, so they move the other way ($i = -nevS$). Their drift speed is tiny compared with the speed of the signal.

An ammeter is connected:

- ☐ in parallel across the component, and has a very large internal resistance;
☒ in series in the branch, and has a very small internal resistance;
☐ in series in the branch, and has a very large internal resistance;
☐ in parallel across the component, and has a very small internal resistance.

Why: the current to be measured must flow *through* it (the circuit is opened and the ammeter inserted), and it must not change the resistance of the branch: $R_A \rightarrow 0$.

Which of these units is one of the seven SI base units?

- ☐ coulomb (C) ☐ volt (V) ☒ ampere (A) ☐ ohm (Ω)

Why: the ampere is a base unit; $\text{C} = \text{A s}$, $\text{V} = \text{W A}^{-1}$ and $\Omega = \text{V A}^{-1}$ are derived units.

Exercise 4. Counting electrons

(5 marks)

A steady current i flows through a wire during a time Δt . Determine the number N of electrons that cross a cross-section of the wire during Δt .

$$i = 3.2 \text{ mA}$$

$$\Delta t = 5.0 \text{ s}$$

$$e = 1.6 \times 10^{-19} \text{ C}$$

Steady current: $q = i \Delta t$ (from $i = dq/dt$ with i constant). Each electron carries a charge of magnitude e , so $q = Ne$:

$$N = \frac{i \Delta t}{e} \quad \text{with } i = 3.2 \text{ mA} = 3.2 \times 10^{-3} \text{ A}$$

$$N = \frac{3.2 \times 10^{-3} \text{ A} \times 5.0 \text{ s}}{1.6 \times 10^{-19} \text{ C}} = \frac{3.2 \times 5.0}{1.6} \times 10^{-3+19} = 10 \times 10^{16}$$

$$N = 1.0 \times 10^{17} \quad N \text{ is a pure number: } \text{A s C}^{-1} = 1.$$

Marking

①	0	1
②	0	1
③	0	1
④	0	1
⑤	0	1

Marking scheme

- ① 1 pt Literal expression $N = i \Delta t / e$, letters only.
- ② 1 pt Conversion $3.2 \text{ mA} = 3.2 \times 10^{-3} \text{ A}$.
- ③ 1 pt Numerical expression written *with units*.
- ④ 1 pt $N = 1.0 \times 10^{17}$.
- ⑤ 1 pt No unit for N (units cancel), two significant figures.

Examiner's tip

Without a calculator, handle the mantissas and the powers of ten separately. In Test 1 the slip was $\frac{1}{3.2 \times 10^4}$: it equals $\frac{1}{3.2} \times 10^{-4} \approx 0.31 \times 10^{-4} = 3.1 \times 10^{-5} \text{ m s}^{-1}$, not 3.2×10^{-3} .

Your score: / 14 → mark out of 20 = score $\times$ 20/14 = / 20

Progress Test 1.C — Elec-S1

1.C · Circuits and two-terminal components

Semester 1 · Academic Year 2026–2027

ANSWER KEY · MODEL ANSWERS AND MARKING SCHEME

Duration: 7 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Unit conversion

(1 mark)

Convert the power $P = 2.5 \times 10^{-2}$ kW into milliwatts. Give the exact result, without rounding.

$P = 2.5 \times 10^{-2} \times 10^3 \text{ W} = 25 \text{ W} = 25 \times 10^3 \text{ mW}$	$P = 2.5 \times 10^4 \text{ mW}$
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Marking

①
0 1

Marking scheme

① 1 pt $2.5 \times 10^4 \text{ mW}$ (or 25 000 mW), unit written correctly.

Exercise 2. Electrical power — definition

(5 marks)

A two-terminal component, studied in the receiver convention, has a voltage $u(t)$ across it and carries a current $i(t)$. Draw the receiver convention and give the power $p(t)$ received by the component, with its SI unit expressed in terms of volts and amperes. Then give the energy it receives between t_1 and t_2 , and its expression in the DC regime ($u = u_0$ and $i = i_0$ constant during Δt).

	Receiver convention: the arrows of u and i point in opposite directions. Power received: $p(t) = u(t) i(t)$ ($p > 0$ for a receiver) Unit: the watt, $1 \text{ W} = 1 \text{ V} \times 1 \text{ A}$. Energy received: $E = \int_{t_1}^{t_2} p(t) dt = \int_{t_1}^{t_2} u(t) i(t) dt$ (in joules) DC regime: $E = u_0 i_0 \Delta t$
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Marking

①
0 1 2

②
0 1 2

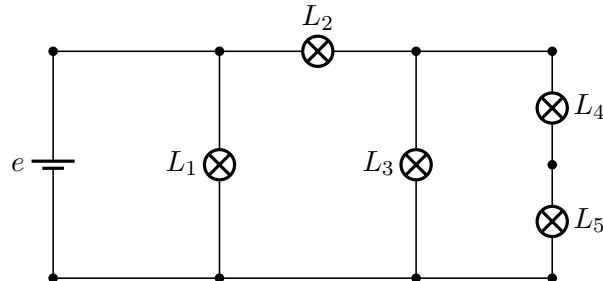
③
0 1

Marking scheme

- ① **2 pts** Receiver convention drawn, arrows opposite (1 pt); $p = u i$ (1 pt).
- ② **2 pts** $E = \int p dt$ (1 pt); $E = u_0 i_0 \Delta t$ in the DC regime (1 pt).
- ③ **1 pt** Watt, $W = V A$.

Exercise 3. Reading a circuit diagram

(4 marks)



Base your answers on the diagram exactly as drawn, even though the same circuit could be drawn with fewer wires.

How many nodes?

☐ 2

☒ 3

☐ 4

☐ 9

Why: a node is a junction of at least three conductors, and all the points joined by wires alone are one and the same node (same potential). Distinct nodes: the top-left wire (e , L_1 , L_2), the top-right wire (L_2 , L_3 , L_4) and the whole bottom wire (e , L_1 , L_3 , L_5). The dot between L_4 and L_5 joins only two conductors, so it is not a node. “4” counts the drawn junction dots: that was the trap in Test 1 (expected answer 2, not 4).

How many branches?

☐ 4

☒ 5

☐ 6

☐ 11

Why: a branch joins two nodes: e , L_1 , L_2 , L_3 , and L_4 with L_5 in series (one single branch, since their common point is not a node).

How many two-terminal components, connecting wires included?

☐ 6

☐ 9

☒ 11

☐ 14

Why: 6 components (e , $L_1, \dots, L_5$) + 5 connecting wires (each plain segment between two dots: 1 top-left, 1 top-right, 3 at the bottom) = 11. In Test 1 the answer was $4 + 6 = 10$ on the same principle.

How many independent meshes (independent mesh equations)?

☐ 2

☒ 3

☐ 4

☐ 5

Why: $m = B - N + 1 = 5 - 3 + 1 = 3$: the three “windows” of the diagram.

Exercise 4. Energy received by a phone

(4 marks)

A phone is charged in the DC regime: the charger applies $u = 5.0 \text{ V}$ and delivers $i = 2.0 \text{ A}$ during $\Delta t = 1 \text{ h } 30 \text{ min}$. Compute the energy received by the phone, in joules, then in watt-hours.

$u = 5.0 \text{ V}$	$i = 2.0 \text{ A}$	$\Delta t = 1 \text{ h } 30 \text{ min}$
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$E = u i \Delta t$ with $\Delta t = 1.5 \text{ h} = 1.5 \times 3600 \text{ s} = 5400 \text{ s}$
$E = 5.0 \text{ V} \times 2.0 \text{ A} \times 5400 \text{ s} = 10 \text{ W} \times 5400 \text{ s}$ $E = 5.4 \times 10^4 \text{ J}$
In watt-hours: $E = 10 \text{ W} \times 1.5 \text{ h}$ $E = 15 \text{ W h}$ (check: $15 \times 3600 = 54000$)

Marking

①	
0	1
②	
0	1
③	
0	1
④	
0	1

Marking scheme

- ① **1 pt** Literal expression $E = u i \Delta t$.
- ② **1 pt** $\Delta t = 5400 \text{ s}$.
- ③ **1 pt** $E = 5.4 \times 10^4 \text{ J}$.
- ④ **1 pt** $E = 15 \text{ W h}$.

Examiner's tip

“1 h 30 min” is 1.5 h, not 1.3 h. Convert times into seconds before using SI formulas.

Your score: / 14 $\longrightarrow$ mark out of 20 = $\text{score} \times 20/14 = \dots\dots\dots / 20$

Progress Test 2.A — Elec-S1

2.A · Characteristic curve and experimental evidence for Ohm's law

Semester 1 · Academic Year 2026–2027

ANSWER KEY · MODEL ANSWERS AND MARKING SCHEME

Duration: 10 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Reading a slope

(1 mark)

The characteristic $i = f(u)$ of a resistor is a straight line through the origin, of slope 20 mA V^{-1} . Give its resistance R .

For $i = f(u)$: $i = \frac{u}{R}$, so the slope is $\frac{1}{R} = G = 20 \times 10^{-3} \text{ A V}^{-1}$
$R = \frac{1}{2.0 \times 10^{-2} \text{ A V}^{-1}}$ $R = 50 \text{ } \Omega$

Marking

①
0 1
☐ ☐

Marking scheme

① 1 pt $R = 50 \text{ } \Omega$. (Answering $R = 20$ or 0.02 confuses the slope with R .)

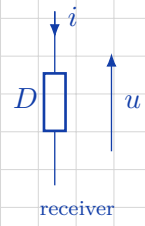
Examiner's tip

On $i = f(u)$ (u on the horizontal axis) the slope is $1/R = G$. On $u = g(i)$ it is R (Fig. 18 of the notes). Always check which quantity is on which axis.

Exercise 2. Characteristic of a component — definition

(5 marks)

Define the characteristic of a two-terminal component. Which convention is used to plot the characteristic of a passive component? Draw it and give the sign of the power received. What is the shape of the characteristic of an ohmic conductor?

The characteristic is the curve $i = f(u)$ giving the current i through the component as a function of the voltage u across it (u on the horizontal axis, i on the vertical axis). It is obtained point by point by measuring pairs (u_k, i_k) for various settings. Passive component: receiver convention. The current enters through the + terminal, so the arrows of u and i are opposite, and $p = ui > 0$ is <i>received</i>. Ohmic conductor: a straight line through the origin (slope $1/R$).	
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Marking

①
0 1 2
☐ ☐ ☐
②
0 1 2
☐ ☐ ☐
③
0 1
☐ ☐

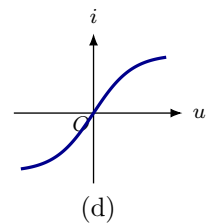
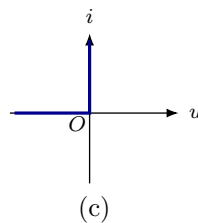
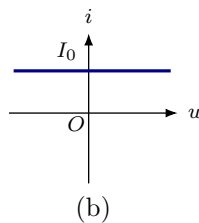
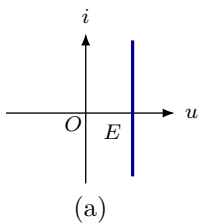
Marking scheme

- ① **2 pts** Curve $i = f(u)$ (1 pt), with the axes stated and measured point by point (1 pt).
- ② **2 pts** Receiver convention drawn with opposite arrows (1 pt); $p = ui > 0$ received (1 pt).
- ③ **1 pt** Straight line through the origin.

Exercise 3. Identifying characteristics

(3 marks)

The four characteristics below are plotted in the (u, i) plane.



Characteristic (a) is that of:

- | | |
|--|---|
| ☐ an ohmic conductor | ☒ an ideal voltage source |
| ☐ an ideal current source | ☐ a lamp |

Why: whatever the current, the voltage keeps the same value $u = E$: this is the definition of an ideal voltage source (vertical line). (b) is an ideal current source, (d) a lamp.

Characteristic (c) is that of:

- | | |
|--|--|
| ☐ a Zener diode | ☐ a lamp |
| ☒ an ideal diode | ☐ an ideal current source |

Why: no current for $u < 0$ (blocked), and $u = 0$ whatever $i > 0$ (conducting). A Zener diode would also conduct in reverse below $-U_Z$.

Which statement is correct?

- ☐ Every characteristic is a straight line through the origin.
- ☒ A straight-line characteristic through the origin is the signature of an ohmic conductor.
- ☐ The characteristic of a lamp is a straight line through the origin.
- ☐ A characteristic does not depend on how u and i are oriented.

Why: "Beware! Not all characteristics are straight lines through the origin" (notes, 2.A). The characteristic does depend on the convention chosen.

Exercise 4. Estimating a resistance from measurements

(5 marks)

The characteristic of a resistor was measured in the receiver convention. The measurements at low current are the noisiest.

i_k (mA)	1.0	2.0	4.0
u_k (V)	0.70	0.90	2.00

(a) Compute the naive estimate $R_{\text{naive}} = \frac{1}{N} \sum_k u_k / i_k$. (b) Compute the least-squares estimate $\hat{R} = \sum_k u_k i_k / \sum_k i_k^2$. (c) Which value should be kept? Justify.

(a) Ratios (V/mA = kΩ): $\frac{0.70 \text{ V}}{1.0 \text{ mA}} = 0.70 \text{ kΩ}$, $\frac{0.90}{2.0} = 0.45 \text{ kΩ}$, $\frac{2.00}{4.0} = 0.50 \text{ kΩ}$
$R_{\text{naive}} = \frac{0.70 + 0.45 + 0.50}{3} = \frac{1.65}{3}$ $R_{\text{naive}} = 0.55 \text{ kΩ} = 550 \text{ Ω}$
(b) $\sum u_k i_k = 0.70 \times 1.0 + 0.90 \times 2.0 + 2.00 \times 4.0 = 10.5 \text{ V mA}$
$\sum i_k^2 = 1.0^2 + 2.0^2 + 4.0^2 = 21 \text{ mA}^2$
$\hat{R} = \frac{10.5}{21} \text{ kΩ}$ $\hat{R} = 0.500 \text{ kΩ} = 500 \text{ Ω}$
(c) Keep $\hat{R} = 500 \text{ Ω}$. The relative uncertainty on u_k / i_k is dominated by that on i_k , which is largest at low current. The naive mean gives the noisy first point (0.70 kΩ) the same weight as the others; the regression weights each point by i_k^2 .

Marking

①	0	1
②	0	1
③	0	1
④	0	1
⑤	0	1

Marking scheme

- ① 1 pt The three ratios, with a correct unit (V/mA = kΩ).
- ② 1 pt $R_{\text{naive}} = 0.55 \text{ kΩ} = 550 \text{ Ω}$.
- ③ 1 pt $\sum u_k i_k = 10.5 \text{ V mA}$ and $\sum i_k^2 = 21 \text{ mA}^2$.
- ④ 1 pt $\hat{R} = 500 \text{ Ω}$.
- ⑤ 1 pt $\hat{R}$ kept, justified (low-current points noisier / weighting by i_k^2).

Examiner's tip

V/mA gives kΩ, not Ω: a factor of 1000 is the most common slip here (Tab. 3 of the notes lists r_k in kΩ). Check: all points should lie close to $u = \hat{R} i$; here the residuals are +0.20, -0.10 and 0 V.

Your score: / 14 → mark out of 20 = score $\times$ 20/14 = / 20

Progress Test 2.B — Elec-S1

2.B · The ohmic conductor and Ohm's law

Semester 1 · Academic Year 2026–2027

ANSWER KEY · MODEL ANSWERS AND MARKING SCHEME

Duration: 7 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Conductance

(1 mark)

A resistor has a conductance $G = 0.25 \text{ mS}$. Give its resistance R in kilohms, exactly.

$R = \frac{1}{G} = \frac{1}{0.25 \times 10^{-3} \text{ S}} = 4.0 \times 10^3 \Omega$	$R = 4.0 \text{ k}\Omega$
--	---------------------------

Marking

①
0 1

Marking scheme

① 1 pt $R = 4.0 \text{ k}\Omega$ (G in siemens, $1 \text{ S} = 1 \Omega^{-1}$).

Exercise 2. Ohm's law

(5 marks)

State Ohm's law for a resistor of resistance R , specifying the convention used (draw it). How does the law read in the other convention? State the unit of R (name and symbol) and express it in terms of volts and amperes.

	Receiver convention (arrows opposite): $u = Ri$		Generator convention (arrows in the same direction): $u = -Ri$
$R > 0$ is the resistance, in ohms (Ω): $1 \Omega = 1 \text{ V A}^{-1}$. (Its reciprocal $G = 1/R$ is in siemens.)			

Marking

①
0 1 2
②
0 1 2
③
0 1

Marking scheme

- ① 2 pts Receiver convention drawn (1 pt) with $u = Ri$ (1 pt).
 ② 2 pts Generator convention: $u = -Ri$ (the sign is the point).
 ③ 1 pt Ohm, $\Omega = \text{V A}^{-1}$.

Examiner's tip

“The formula for Ohm’s law changes according to the convention chosen” (notes, remark v). Draw the arrows *before* writing $u = \pm Ri$.

Exercise 3. Resistances in practice

(3 marks)

The typical input resistance of a digital voltmeter is:

- ☐ a few m Ω ☐ a few Ω ☐ a few k Ω ☒ 1 to 10 M Ω

Why: a voltmeter must draw almost no current, so R_V is very large (Tab. 6 of the notes).

To measure, with an ohmmeter, a resistor that is part of a circuit, you must:

- ☐ leave the circuit powered, so that a current flows through the resistor;
☐ switch the power supply off, but leave the resistor connected;
☒ disconnect the power supply and at least one terminal of the resistor;
☐ connect the ohmmeter in series with the resistor.

Why: the ohmmeter injects its own small current; any other source or parallel path would distort the reading. Disconnect, do not merely switch off.

A resistor R carries a current i . The power dissipated by the Joule effect is:

- ☐ Ri ☐ R^2i ☒ Ri^2 ☐ u^2R

Why: $p = ui$ with $u = Ri$ gives $p = Ri^2 = u^2/R$. Check the dimensions: $\Omega \times A^2 = V A = W$.

Exercise 4. Joule effect — a rear-window demister

(5 marks)

A car’s rear-window demister is modelled by an ohmic resistor supplied by the car battery in the DC regime. Determine its resistance R , the current i it carries, then the energy E dissipated as heat in $\Delta t = 10$ min.

$u = 12 \text{ V}$	$p = 72 \text{ W}$	$\Delta t = 10 \text{ min}$
--------------------	--------------------	-----------------------------

$$p = \frac{u^2}{R} \Rightarrow R = \frac{u^2}{p} = \frac{(12 \text{ V})^2}{72 \text{ W}} = \frac{144}{72} \Omega \quad R = 2.0 \Omega$$

$$i = \frac{u}{R} = \frac{12 \text{ V}}{2.0 \Omega} \quad i = 6.0 \text{ A} \quad (\text{check: } ui = 12 \times 6.0 = 72 \text{ W})$$

$$E = p \Delta t \text{ with } \Delta t = 10 \times 60 = 600 \text{ s: } E = 72 \text{ W} \times 600 \text{ s}$$

$$E = 4.3 \times 10^4 \text{ J} \quad (43.2 \text{ kJ})$$

Marking

①	
0	1
②	
0	1
③	
0	1
④	
0	1
⑤	
0	1

Marking scheme

- ① 1 pt Literal expression $R = u^2/p$, obtained from $p = u^2/R$.
 ② 1 pt $R = 2.0 \Omega$.
 ③ 1 pt $i = 6.0 \text{ A}$ (from u/R or p/u).
 ④ 1 pt $\Delta t = 600 \text{ s}$.
 ⑤ 1 pt $E = 4.3 \times 10^4 \text{ J}$, with its unit.

Your score: / 14 $\rightarrow$ mark out of 20 = score $\times 20/14 =$ / 20

Progress Test 2.C — Elec-S1

2.C · Pouillet's law

Semester 1 · Academic Year 2026–2027

ANSWER KEY · MODEL ANSWERS AND MARKING SCHEME

Duration: 7 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Unit conversion

(1 mark)

Tables often give resistivities in $\mu\Omega\cdot\text{cm}$. Convert $\rho = 1.7 \mu\Omega\cdot\text{cm}$ into $\Omega\cdot\text{m}$. Give the exact result.

$\rho = 1.7 \times 10^{-6} \Omega \times 10^{-2} \text{ m}$	$\rho = 1.7 \times 10^{-8} \Omega \text{ m}$	(the resistivity of copper)
---	--	-----------------------------

Marking

①
0 1

Marking scheme

① 1 pt $1.7 \times 10^{-8} \Omega\cdot\text{m}$.

Exercise 2. Pouillet's law

(5 marks)

State Pouillet's law giving the resistance R of a conducting wire of length L and cross-sectional area S , made of a material of resistivity ρ . Give the SI unit of each of the four quantities. Define the conductivity σ and give its unit.

$R = \rho \frac{L}{S}$	a longer wire resists more, a thicker wire conducts more easily.
Units: R in ohms (Ω), ρ in $\Omega\cdot\text{m}$, L in metres (m), S in square metres (m^2). (check: $\Omega\cdot\text{m} \times \text{m}/\text{m}^2 = \Omega$)	
Conductivity: $\sigma = \frac{1}{\rho}$, in siemens per metre ($\text{S}\cdot\text{m}^{-1}$).	

Marking

①
0 1 2
②
0 1 2
③
0 1

Marking scheme

- ① 2 pts $R = \rho L/S$ (0 if L and S are swapped).
- ② 2 pts The four units (1 pt lost per mistake).
- ③ 1 pt $\sigma = 1/\rho$ in $\text{S}\cdot\text{m}^{-1}$.

Examiner's tip

Checking the dimensions ($\Omega \text{ m} \times \text{m}/\text{m}^2 = \Omega$) catches a swapped L and S in two seconds.

Exercise 3. Resistivity and geometry

(3 marks)

A wire is replaced by a wire of the same material, twice as long and with twice the diameter. Its resistance is:

- ☐ multiplied by 2 ☐ unchanged ☒ divided by 2 ☐ divided by 4

Why: $L \rightarrow 2L$, and doubling the diameter multiplies $S = \pi d^2/4$ by 4, so $R \rightarrow R \times 2/4 = R/2$.

Among these materials, the best conductor is:

- ☐ iron ☐ aluminium ☐ copper ☒ silver

Why: it has the lowest resistivity: $\rho_{\text{Ag}} = 1.6 \times 10^{-8} \Omega \text{ m}$ (Cu: 1.7, Al: 2.8, Fe: 10×10^{-8}).

When a metallic wire heats up, its resistance:

- ☒ increases ☐ decreases ☐ stays the same ☐ vanishes

Why: for most metals the resistivity increases with temperature: a heated metal conducts slightly less well.

Exercise 4. Resistance of a copper wire

(5 marks)

A copper wire of length L and cross-sectional area S carries a current i . Determine its resistance R , then the power p it dissipates by the Joule effect.

$$L = 20 \text{ m} \quad S = 0.50 \text{ mm}^2 \quad \rho = 1.7 \times 10^{-8} \Omega \text{ m} \quad i = 5.0 \text{ A}$$

$$R = \rho \frac{L}{S} \quad \text{with } S = 0.50 \text{ mm}^2 = 0.50 \times (10^{-3} \text{ m})^2 = 5.0 \times 10^{-7} \text{ m}^2$$

$$R = \frac{1.7 \times 10^{-8} \Omega \text{ m} \times 20 \text{ m}}{5.0 \times 10^{-7} \text{ m}^2} = \frac{1.7 \times 20}{5.0} \times 10^{-8+7} \Omega = 6.8 \times 10^{-1} \Omega$$

$$R = 0.68 \Omega$$

$$p = Ri^2 = 0.68 \Omega \times (5.0 \text{ A})^2 = 0.68 \times 25 \text{ W} \quad p = 17 \text{ W}$$

Marking

①
0 1
☐
②
0 1
☐
③
0 1
☐
④
0 1
☐
⑤
0 1
☐

Marking scheme

- ① 1 pt Literal expression $R = \rho L/S$.
 ② 1 pt $S = 5.0 \times 10^{-7} \text{ m}^2$ ($1 \text{ mm}^2 = 10^{-6} \text{ m}^2$).
 ③ 1 pt $R = 0.68 \Omega$.
 ④ 1 pt Literal expression $p = Ri^2$.
 ⑤ 1 pt $p = 17 \text{ W}$, with its unit.

Examiner's tip

$\text{mm}^2 \rightarrow \text{m}^2$: square the conversion factor, $(10^{-3})^2 = 10^{-6}$. You got this right in Test 1; keep writing it out explicitly.

Your score: / 14 $\rightarrow$ mark out of 20 = score $\times$ 20/14 = / 20

Progress Test 3.A — Elec-S1

3.A · Kirchhoff's laws (combinations, divider, Millman)

Semester 1 · Academic Year 2026–2027

ANSWER KEY · MODEL ANSWERS AND MARKING SCHEME

Duration: 10 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Equivalent resistance

(1 mark)

Give the equivalent resistance of $R_1 = 6.0 \text{ k}\Omega$ and $R_2 = 3.0 \text{ k}\Omega$ connected in parallel.

$R_{\text{eq}} = \frac{R_1 R_2}{R_1 + R_2} = \frac{6.0 \times 3.0}{6.0 + 3.0} \text{ k}\Omega = \frac{18}{9.0} \text{ k}\Omega$	$R_{\text{eq}} = 2.0 \text{ k}\Omega$	(smaller than the smaller one)
---	---------------------------------------	--------------------------------

Marking

①
0 1

Marking scheme

① 1 pt $R_{\text{eq}} = 2.0 \text{ k}\Omega$ (“product over sum”).

Exercise 2. Kirchhoff's laws

(5 marks)

State Kirchhoff's node law and mesh law. For each law, give the fundamental principle it expresses.

Node law. At any node, the sum of the currents entering the node equals the sum of the currents leaving it: $\sum i_{\text{in}} = \sum i_{\text{out}}$, i.e. $\sum_k \varepsilon_k i_k = 0$ with $\varepsilon_k = +1$ for a current entering and -1 for a current leaving. Mesh law. Along any mesh (closed path), the algebraic sum of the voltages is zero: $\sum_k \varepsilon_k u_k = 0$, with $\varepsilon_k = +1$ if the arrow of u_k points in the direction of travel and -1 otherwise. The node law expresses the conservation of electric charge (no charge piles up at a node). The mesh law expresses the conservation of energy (after a closed path the potential returns to its starting value).
--

Marking

①
0 1 2
②
0 1 2
③
0 1

Marking scheme

- ① 2 pts Node law: statement (1 pt) and algebraic form with signs (1 pt).
 ② 2 pts Mesh law: statement (1 pt) and sign convention along the path (1 pt).
 ③ 1 pt Charge conservation *and* energy conservation, each matched to its law.

Examiner's tip

Box 6 of the handout is easy to misread: its brackets refer to the *first* sentence of each paragraph. The same current through components in series comes from the node law; the same voltage across components in parallel comes from the mesh law. Currents add up at a node (node law); voltages add up round a loop (mesh law).

Exercise 3. Quick circuit laws

(3 marks)

$R_1 = 1.0 \text{ k}\Omega$ and $R_2 = 3.0 \text{ k}\Omega$ are in series under $u = 12 \text{ V}$. The voltage u_2 across R_2 is:

- ☐ 3.0 V ☐ 4.0 V ☒ 9.0 V ☐ 12 V

Why: voltage divider: $u_2 = \frac{R_2}{R_1 + R_2} u = \frac{3}{4} \times 12 = 9.0 \text{ V}$. The larger resistance takes the larger share.

A circuit has $N = 3$ electrically distinct nodes and $B = 5$ branches. The number of independent mesh equations is:

- ☐ 2 ☒ 3 ☐ 5 ☐ 8

Why: $m = B - N + 1 = 5 - 3 + 1 = 3$ (and there are $N - 1 = 2$ independent node equations: $2 + 3 = 5$ equations for the 5 unknown branch currents).

Three identical 30Ω resistors in parallel are equivalent to:

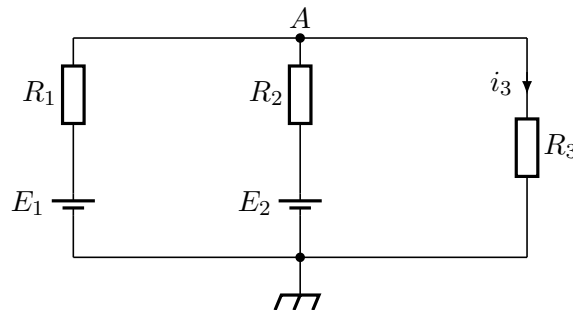
- ☐ 90 Ω ☐ 30 Ω ☒ 10 Ω ☐ 3.3 Ω

Why: $1/R_{\text{eq}} = 3/30 \text{ S}$, so $R_{\text{eq}} = 10 \Omega$.

Exercise 4. Millman's theorem

(5 marks)

In the circuit below, the ground (bottom wire) is the reference of potentials. Using Millman's theorem, determine the potential V_A of node A. Deduce the current i_3 that flows through R_3 towards the ground.



$$E_1 = 12 \text{ V} \quad R_1 = 2.0 \Omega \quad E_2 = 6.0 \text{ V} \quad R_2 = 3.0 \Omega \quad R_3 = 6.0 \Omega$$

The potentials at the far ends of the three branches are E_1 , E_2 and 0:

$$V_A = \frac{\frac{E_1}{R_1} + \frac{E_2}{R_2} + \frac{0}{R_3}}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}$$

$$\text{Numerator: } \frac{12}{2.0} + \frac{6.0}{3.0} + 0 = 6.0 + 2.0 = 8.0 \text{ A} \quad \text{Denominator: } \frac{1}{2} + \frac{1}{3} + \frac{1}{6} = \frac{3+2+1}{6} = 1.0 \text{ S}$$

$$V_A = 8.0 \text{ V} \quad i_3 = \frac{V_A - 0}{R_3} = \frac{8.0}{6.0} \text{ A} \quad i_3 = 1.3 \text{ A} \quad (4/3 \text{ A})$$

$$\text{Check (node law at A): } i_1 = \frac{E_1 - V_A}{R_1} = 2.0 \text{ A enters A, } \frac{E_2 - V_A}{R_2} = -0.67 \text{ A, and } 2.0 - 0.67 = 1.33 \text{ A} = i_3. \checkmark$$

Marking

①	
0	1
②	
0	1
③	
0	1
④	
0	1
⑤	
0	1

Marking scheme

- ① **1 pt** Millman's formula written literally, with the right potentials (E_1 , E_2 , 0).
- ② **1 pt** Numerator = 8.0 A.
- ③ **1 pt** Denominator = 1.0 S.
- ④ **1 pt** $V_A = 8.0$ V.
- ⑤ **1 pt** $i_3 = V_A/R_3 = 1.3$ A.

Examiner's tip

Millman gives V_A as a mean of the potentials weighted by the conductances. The result must lie between the smallest and the largest potential (0 and 12 V): a quick sanity check.

Your score: / 14 $\longrightarrow$ mark out of 20 = score $\times$ 20/14 = / 20

Progress Test 3.B — Elec-S1
3.B · Thévenin’s and Norton’s equivalence theorems
Semester 1 · Academic Year 2026–2027

ANSWER KEY · MODEL ANSWERS AND MARKING SCHEME

Duration: 10 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. From Norton to Thévenin (1 mark)

A Norton model has $I_{No} = 2.0\text{mA}$ and $R_{No} = 4.7\text{k}\Omega$. Give the electromotive force E_{Th} of the equivalent Thévenin model.

$E_{Th} = R I_{No} = 4.7 \times 10^3 \, \Omega \times 2.0 \times 10^{-3} \, \text{A}$	$E_{Th} = 9.4 \, \text{V}$	(with
$R_{Th} = R_{No} = 4.7 \, \text{k}\Omega)$		

Marking

①
0 1
☐ ☐

Marking scheme
① 1 pt $E_{Th} = 9.4 \, \text{V}$ ($\text{k}\Omega \times \text{mA} = \text{V}$).

Exercise 2. Thévenin's theorem

(5 marks)

State Thévenin's theorem for a network made of ideal sources and ohmic conductors, seen from its terminals A and B (draw the model). Explain how E_{Th} and R_{Th} are determined.

Seen from A and B , the network is equivalent to a single ideal voltage source E_{Th} in series with a single resistor R_{Th} :	
$u = E_{Th} - R_{Th} i$ (generator convention).	
E_{Th} = open-circuit voltage between A and B ($i = 0$), with the sources left active (use the divider, Kirchhoff, ...).	
R_{Th} = resistance seen between A and B with all sources switched off : ideal voltage sources replaced by short circuits, ideal current sources by open circuits.	

Marking

①		
0	1	2

②		
0	1	2

③	
0	1

Marking scheme

- ① **2 pts** Model drawn: E_{Th} in series with R_{Th} (1 pt); $u = E_{Th} - R_{Th} i$ or an equivalent statement (1 pt).
 ② **2 pts** E_{Th} = open-circuit voltage (1 pt), sources active and $i = 0$ (1 pt).
 ③ **1 pt** R_{Th} with the sources off: voltage sources shorted, current sources opened.

Exercise 3. Thévenin and Norton models

(3 marks)

To switch off an ideal voltage source, it is replaced by:

- ☐ an open circuit ☒ a short circuit (a wire)
☐ a resistor R_{Th} ☐ an ideal current source

Why: a switched-off voltage source imposes $u = 0$ whatever the current: that is a wire. A switched-off current source imposes $i = 0$: that is an open circuit.

A Thévenin model has $E_{Th} = 12\text{ V}$ and $R_{Th} = 3.0\ \Omega$. The equivalent Norton model has:

- ☐ $I_{No} = 36\text{ A}$, $R_{No} = 3.0\ \Omega$ ☒ $I_{No} = 4.0\text{ A}$, $R_{No} = 3.0\ \Omega$
☐ $I_{No} = 4.0\text{ A}$, $R_{No} = 0.33\ \Omega$ ☐ $I_{No} = 0.25\text{ A}$, $R_{No} = 3.0\ \Omega$

Why: $I_{No} = E_{Th}/R = 12/3.0 = 4.0\text{ A}$, and the resistance is common to both models.

A Thévenin (or Norton) model can describe a “black box” containing:

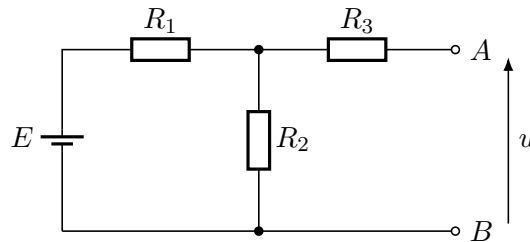
- ☐ a diode;
☒ only ideal sources and ohmic conductors;
☐ a lamp;
☐ a motor.

Why: the models are valid only for ideal sources and ohmic conductors: “no lamp, no motor, no diode” (notes, 3.B).

Exercise 4. Thévenin model of a network

(5 marks)

Determine the Thévenin model (E_{Th} , R_{Th}) of the network below, seen from the terminals A and B . Deduce the current I_{No} of the equivalent Norton model.



$$E = 12 \text{ V}$$

$$R_1 = 6.0 \text{ k}\Omega$$

$$R_2 = 3.0 \text{ k}\Omega$$

$$R_3 = 1.0 \text{ k}\Omega$$

Sources off (E replaced by a wire): R_1 and R_2 are then in parallel, and this pair is in series with R_3 :

$$R_{Th} = R_3 + \frac{R_1 R_2}{R_1 + R_2} = 1.0 + \frac{18}{9.0} = 1.0 + 2.0 \quad \boxed{R_{Th} = 3.0 \text{ k}\Omega}$$

Open circuit ($i = 0$): no current in R_3 , so no voltage across it and u_{AB} equals the voltage across R_2 (voltage divider):

$$E_{Th} = \frac{R_2}{R_1 + R_2} E = \frac{3.0}{9.0} \times 12 \text{ V} \quad \boxed{E_{Th} = 4.0 \text{ V}}$$

$$I_{No} = \frac{E_{Th}}{R_{Th}} = \frac{4.0 \text{ V}}{3.0 \text{ k}\Omega} \quad \boxed{I_{No} = 1.3 \text{ mA}} \quad (4/3 \text{ mA})$$

Marking

①	0	1
②	0	1
③	0	1
④	0	1
⑤	0	1

Marking scheme

- ① 1 pt Sources off: E short-circuited, $R_{Th} = R_3 + R_1 \parallel R_2$.
- ② 1 pt $R_{Th} = 3.0 \text{ k}\Omega$.
- ③ 1 pt Open circuit: no current in R_3 , so a voltage divider across R_2 .
- ④ 1 pt $E_{Th} = 4.0 \text{ V}$.
- ⑤ 1 pt $I_{No} = E_{Th}/R_{Th} = 1.3 \text{ mA}$.

Examiner's tip

R_3 changes R_{Th} but not E_{Th} : in open circuit it carries no current. Check I_{No} directly: with A and B short-circuited, $R_2 \parallel R_3 = 0.75 \text{ k}\Omega$, so E delivers $12/6.75 = 16/9 \text{ mA}$, of which R_3 takes $\frac{R_2}{R_2 + R_3} = \frac{3}{4}$: $I_{No} = \frac{3}{4} \times \frac{16}{9} = \frac{4}{3} \text{ mA}$. ✓

Your score: / 14 → mark out of 20 = score $\times$ 20/14 = / 20

Progress Test 3.C — Elec-S1

3.C · Operating point of a circuit

Semester 1 · Academic Year 2026–2027

ANSWER KEY · MODEL ANSWERS AND MARKING SCHEME

Duration: 10 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Maximum power

(1 mark)

A source has $E_{Th} = 12\text{ V}$ and $R_{Th} = 6.0\ \Omega$. Give the maximum power it can deliver to a resistive load.

Matching $R_c = R_{Th}$:	$P_c^{\max} = \frac{E_{Th}^2}{4R_{Th}} = \frac{144}{4 \times 6.0}\text{ W}$	$P_c^{\max} = 6.0\text{ W}$
---------------------------	---	-----------------------------

Marking

①
0 1

Marking scheme

① 1 pt $P_c^{\max} = 6.0\text{ W}$.

Exercise 2. The operating point

(5 marks)

A linear network, described by its Thévenin model (E_{Th}, R_{Th}), feeds a load resistor R_c . Define the operating point. Write down the characteristic of the source and that of the load, and deduce the analytical expressions of u^* and i^* .

The operating point is the actual state of the circuit: the unique pair (u^*, i^*) that satisfies <i>simultaneously</i> the characteristic of the source and that of the load. Graphically, it is the intersection of the two characteristics. Source (generator convention): $u = E_{Th} - R_{Th} i$ Load (receiver convention): $u = R_c i$	
$R_c i^* = E_{Th} - R_{Th} i^* \Rightarrow i^* = \frac{E_{Th}}{R_{Th} + R_c}$	$u^* = R_c i^* = \frac{R_c}{R_{Th} + R_c} E_{Th}$

Marking

①
0 1 2
②
0 1 2
③
0 1

Marking scheme

- ① **2 pts** Pair (u^*, i^*) satisfying both characteristics (1 pt); intersection of the two curves (1 pt).
- ② **2 pts** $u = E_{Th} - R_{Th}i$ (1 pt) and $u = R_c i$ (1 pt).
- ③ **1 pt** Both i^* and u^* .

Exercise 3. Power transfer

(3 marks)

When the load resistance tends to infinity (open circuit):

- | | |
|---|--|
| ☐ $u^* \rightarrow 0$ and $i^* \rightarrow E_{Th}/R_{Th}$ | ☒ $u^* \rightarrow E_{Th}$ and $i^* \rightarrow 0$ |
| ☐ $u^* \rightarrow E_{Th}$ and $i^* \rightarrow E_{Th}/R_{Th}$ | ☐ $u^* \rightarrow 0$ and $i^* \rightarrow 0$ |

Why: $i^* = E_{Th}/(R_{Th} + R_c) \rightarrow 0$ and $u^* \rightarrow E_{Th}$: the point reaches the voltage axis, and the load receives no power.

When the load is matched to the source ($R_c = R_{Th}$), the efficiency η of the transfer is:

- | | | | |
|--------------------------------|-------------------------------|--|-------------------------------|
| ☐ 100 % | ☐ 75 % | ☒ 50 % | ☐ 25 % |
|--------------------------------|-------------------------------|--|-------------------------------|

Why: $\eta = R_c/(R_{Th} + R_c) = 1/2$: as much power is lost in R_{Th} as reaches the load.

To transmit electric power over long distances, one chooses:

- ☐ matching, $R_c = R_{Th}$, to obtain the maximum power;
- ☒ a high efficiency, $R_c \gg R_{Th}$, to limit the losses;
- ☐ $R_c \ll R_{Th}$, to obtain the maximum current;
- ☐ a short circuit at the end of the line.

Why: energy is costly and losses cause heating, so efficiency wins. Matching is for tiny signals (antenna, sensor), where extracting the most power is what matters.

Exercise 4. Operating point, power and efficiency

(5 marks)

A source with $E_{Th} = 12\text{ V}$ and $R_{Th} = 4.0\ \Omega$ feeds a load $R_c = 8.0\ \Omega$. Determine the operating point (u^*, i^*), the power P_c received by the load and the efficiency η of the transfer. Check the power balance.

$E_{Th} = 12\text{ V}$	$R_{Th} = 4.0\ \Omega$	$R_c = 8.0\ \Omega$
------------------------	------------------------	---------------------

$i^* = \frac{E_{Th}}{R_{Th} + R_c} = \frac{12\text{ V}}{12\ \Omega}$	$i^* = 1.0\text{ A}$	$u^* = R_c i^*$	$u^* = 8.0\text{ V}$
$P_c = R_c i^{*2} = 8.0\ \Omega \times (1.0\text{ A})^2$	$P_c = 8.0\text{ W}$		
$\eta = \frac{P_c}{E_{Th} i^*} = \frac{R_c}{R_{Th} + R_c} = \frac{8.0}{12}$	$\eta = 2/3 \approx 67\%$		
Balance: produced by the ideal source $E_{Th} i^* = 12\text{ W}$ = lost in R_{Th} , $R_{Th} i^{*2} = 4.0\text{ W}$, + received by the load, 8.0 W . ✓			

Marking

①	0	1
	☐	☐
②	0	1
	☐	☐
③	0	1
	☐	☐
④	0	1
	☐	☐
⑤	0	1
	☐	☐

Marking scheme

- ① 1 pt $i^* = 1.0\text{ A}$.
- ② 1 pt $u^* = 8.0\text{ V}$.
- ③ 1 pt $P_c = 8.0\text{ W}$.
- ④ 1 pt $\eta = 2/3 \approx 67\%$.
- ⑤ 1 pt Balance $12\text{ W} = 4.0\text{ W} + 8.0\text{ W}$.

Examiner's tip

η is the ratio of the power received by the load to the power produced by the *ideal* source $E_{Th} i^*$, not to $P_c^{\max}$.

Your score: / 14 $\longrightarrow$ mark out of 20 = score $\times$ 20/14 = / 20